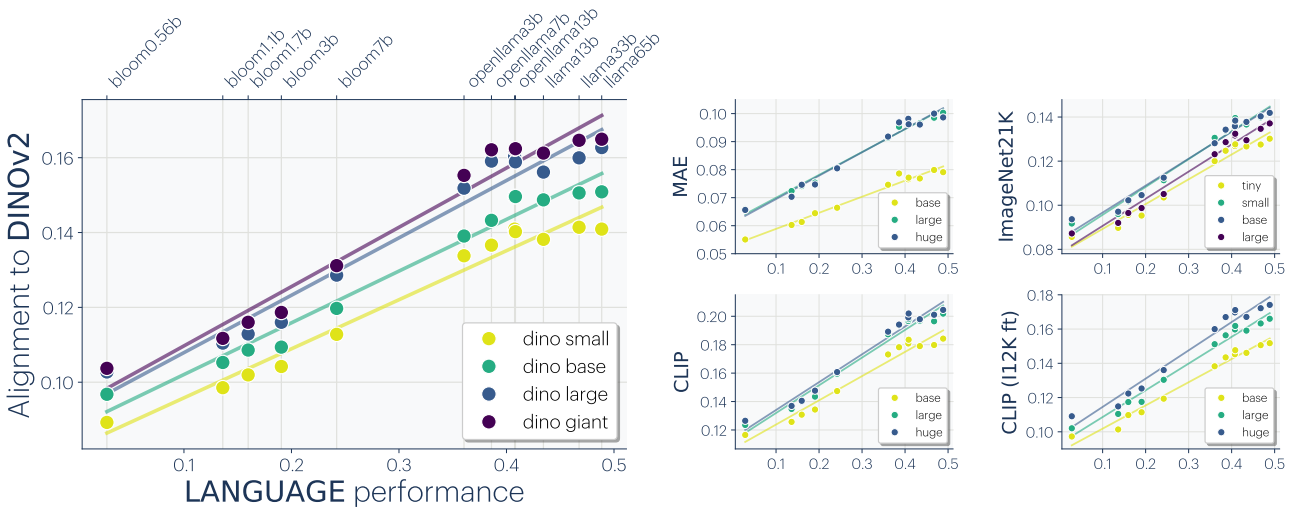

Figure 3. **LANGUAGE and VISION models align:** We measure alignment using mutual nearest-neighbor on the Wikipedia caption dataset (WIT) (Srinivasan et al., 2021). The x-axis is the language model performance measured over 4M tokens from the OpenWebText dataset (Gokaslan & Cohen, 2019) (see Appendix B for plots with model names). We measure performance using $1 - \texttt{bits-per-byte}$, where `bits-per-byte` normalizes the cross-entropy by the total bytes in the input text string. The results show a linear relationship between language-vision alignment and language modeling score, where a general trend is that more capable language models align better with more capable vision models. We find that CLIP models, which are trained with explicit language supervision, exhibit a higher level of alignment. However, this alignment decreases after being fine-tuned on ImageNet classification (labeled CLIP (I12K ft)).

et al., 2022). Because of this, it is possible to merge separately trained models of the same architecture, and achieve some of the capabilities of all models in the mixture (Stoica et al., 2023; Jordan et al., 2022; Wortsman et al., 2022).

2.3. Representations are converging across modalities

Do models trained on different data modalities also converge? Several works indicate that the answer is *yes*.

Merullo et al. (2022) extended model stitching to the cross-modal setting, finding that a single linear projection is sufficient to stitch a vision model to an LLM and achieve good performance on visual question answering and image captioning. Koh et al. (2023) showed that linear stitching can also work in the opposite direction, aligning text inputs to visual outputs. In fact, many recent language-vision models stitch pre-trained language and vision models together. For example, LLaVA (Liu et al., 2023) demonstrated state-of-the-art results by projecting visual features into a language model with a 2-layer MLP.

Other works show further kinds of evidence of cross-modal synergy. OpenAI (2023) found that jointly training a language model with a vision model improves performance on language tasks, compared to training the language model on its own. Sorscher et al. (2022) show a setting in which word embeddings of visual concept names can be isometrically mapped to image embeddings for those same concepts. In work concurrent to ours, Maniparambil et al. (2024) show well-trained vision encoders on large datasets exhibit high semantic similarity with language encoders regardless of the training paradigm (supervised, self-supervised, or language-supervised). Sharma et al. (2024) probed the visual knowledge of LLMs trained *only* on language data, by converting images into code that an LLM can process. They found that LLMs have rich knowledge of visual structures, to the extent that decent visual representations can be trained on images generated solely by querying an LLM to produce code and rendering the response. In visual generation, LLMs show abilities to augment captions with visual structures (*e.g.*, bounding boxes) and improve generation quality (Betker et al., 2023; Lian et al., 2023a;b; Wu et al., 2023). Over other modalities, Ngo & Kim (2024) showed auditory models are also roughly aligned with LLMs up to a linear transformation, and Ng et al. (2023) demonstrated the effectiveness of using pre-trained LLMs for facial motion prediction.

We set out to address these claims in a broader scope to determine whether models are indeed learning an increasingly modality-agnostic representation of the world. We sampled a variety of models trained either solely on vision or solely on language, and compared their representations as they became larger and more competent over many tasks.

In Figure 3, we assess alignment between a suite of language models and vision models. So far we have only defined alignment for two kernels defined over the same input space. To measure cross-modal alignment, we use paired datasets to bridge the two modalities. For vision and text, we use the Wikipedia captions dataset $\{(x_i, y_i)\}_i$ (Srinivasan et al., 2021), composed of images from Wikipedia (x_i) and their

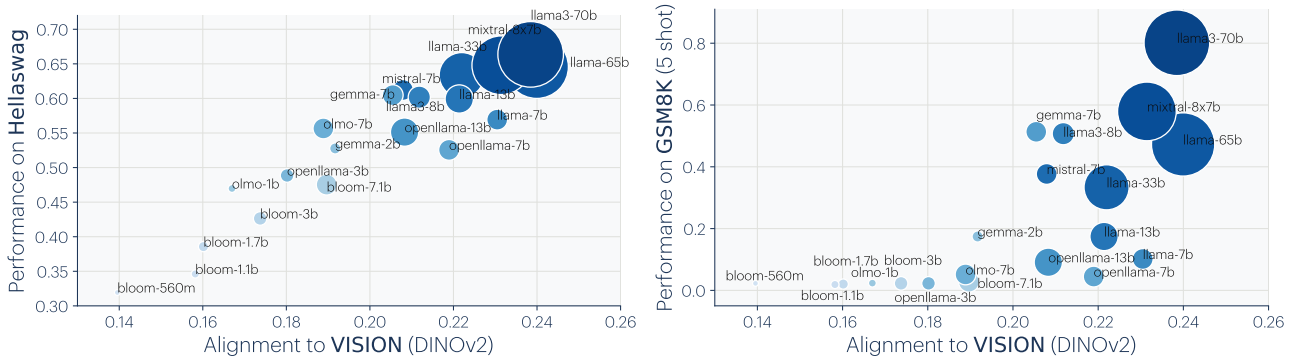

Figure 4. **Alignment predicts downstream performance:** We visualize correlation between LLM alignment score to DINOv2 (Oquab et al., 2023) and downstream task performance on Hellaswag (common-sense) (Zellers et al., 2019) and GSM8K (math) (Cobbe et al., 2021). LLMs are plotted with radii proportional to the size of the model, and color-coded by their rank order in language modeling scores ($1 - \texttt{bits-per-byte}$). We observe that models aligned more closely with vision also show better performance on downstream language tasks. For Hellaswag, there is a linear relationship with alignment score, while GSM8K exhibits an "emergence"-esque trend.

corresponding captions (y_i). We then measure alignment between a language model $f_{\texttt{text}}$ and a vision model $f_{\texttt{img}}$ as the alignment of the two following kernels:

$$K_{\texttt{img}}(i,j) = \langle f_{\texttt{img}}(x_i), f_{\texttt{img}}(x_j) \rangle \quad (1)$$

$$K_{\texttt{text}}(i,j) = \langle f_{\texttt{text}}(y_i), f_{\texttt{text}}(y_j) \rangle. \quad (2)$$

Using this analysis, we find that the better an LLM is at language modeling, the more it tends to aligns with vision models, as shown in Figure 3. The converse effect also holds: the better a vision models is, the more it tends to align with LLMs. See Appendix C.2 for more details.

2.4. Models are increasingly aligning to brains

Neural networks also show substantial alignment with biological representations in the brain (Yamins et al., 2014). This commonality may be due to similarities in the task and data constraints both systems are confronted with. Even though the mediums may differ – silicon transistors versus biological neurons – the fundamental problem faced by brains and machines is the same: efficiently extracting and understanding the underlying structure in images, text, sounds, *etc.* (Barlow et al., 1961; Olshausen & Field, 1997). Sorscher et al. (2022) developed a theoretical framework for how the efficient extraction of novel concepts occurs for both the human visual system and deep networks. The tasks that the human visual system has been honed to perform through evolution – like segmentation, detection, and whole-image classification – are also the ones that we train our neural nets to perform. Yamins et al. (2014) went as far as to title their work in the spirit that performance over such tasks implies brain alignment. Antonello & Huth (2024) posited that it is less the particular task and more the generality of the representations that explain their alignment with biological representations. Further, Conwell et al. (2022) showed that training data plays a large role in alignment. Psychophysical studies have also shown agreement between how humans perceive visual similarity and how models do, even when the models are trained on tasks, such as self-supervised prediction, that are seemingly unrelated to mimicking human perception (Zhang et al., 2018).

2.5. Does alignment predict downstream performance?

If models are converging towards a more accurate representation of reality, we expect that alignment should correspond to improved performance on downstream tasks. Figure 4 supports this hypothesis by demonstrating improved performance on commonsense reasoning (Hellaswag; Zellers et al. (2019)) and mathematical problem solving (GSM8K; Cobbe et al. (2021)) as alignment improves.

3. Why are representations converging?

Modern machine learning models are generally trained to minimize the empirical risk with possible implicit and/or explicit regularization:

$$\underbrace{f^*}_{\text{trained model}} = \arg\min_{f \in \underbrace{\mathcal{F}}_{\text{function class}}} \mathbb{E}_{x \sim \text{dataset}} [\underbrace{\mathcal{L}}_{\text{training objective}}(f, x)] + \underbrace{\mathcal{R}}_{\text{regularization}}(f)$$

In the following sections, we lay out how each colored component in this optimization process potentially plays a role in facilitating representational convergence.

3.1. Convergence via Task Generality

Each training datapoint and objective (task) places an additional constraint on the model. As data and tasks scale, the volume of representations that satisfy these constraints must proportionately grow smaller, as visualized in Figure 6 and stated below:

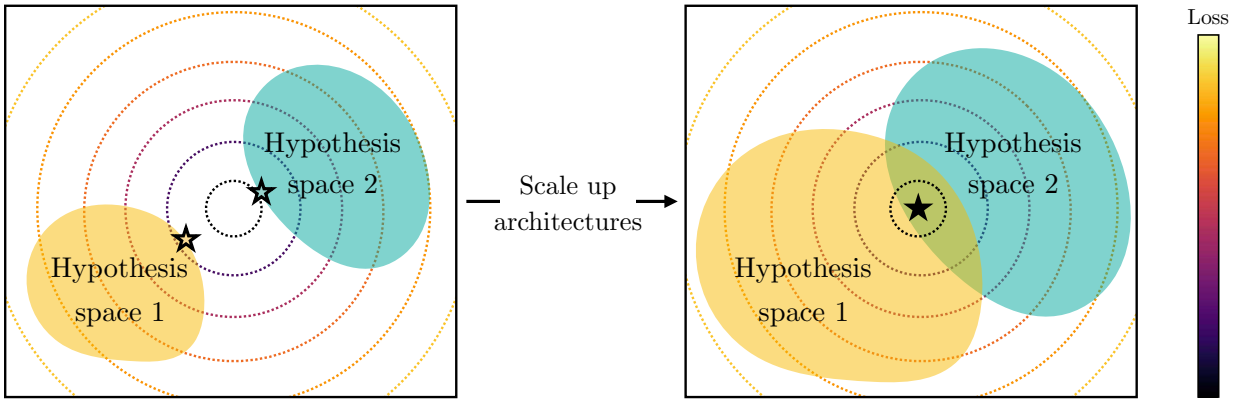

Figure 5. **The Capacity Hypothesis:** If an optimal representation exists in function space, larger hypothesis spaces are more likely to cover it. **LEFT:** Two small models might not cover the optimum and thus find *different* solutions (marked by outlined ☆). **RIGHT:** As the models become larger, they cover the optimum and converge to the same solution (marked by filled ★).

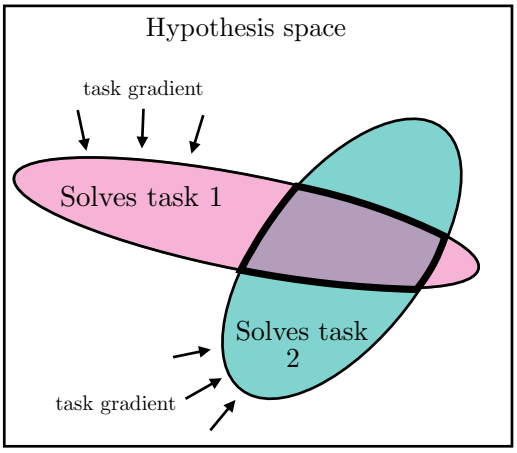

Figure 6. **The Multitask Scaling Hypothesis:** Models trained with an increasing number of tasks are subjected to pressure to learn a representation that can solve all the tasks.

The Multitask Scaling Hypothesis

There are fewer representations that are competent for N tasks than there are for $M < N$ tasks. As we train more general models that solve more tasks at once, we should expect fewer possible solutions.

This has been previously termed as the Contravariance principle by Cao & Yamins (2024), which states that the set of solutions to an easy goal is large, while the set of solutions to a challenging goal is comparatively smaller. Moreover, we argue that this narrower solution set also generalizes better. As data scales, models that optimize the empirical risk $\mathbb{E}_{x\sim \text{dataset}}[\mathcal{L}(f,x)]$ also improve on the population risk $\mathbb{E}_{x\sim \text{reality}}[\mathcal{L}(f,x)]$, and become better at capturing statistical structures of the true data generating process (reality).

Recent work has demonstrated a power law relationship between data scale and model performance (Hestness et al., 2017). This implies that with enough data (*e.g.*, consisting of the entire internet and all offline scientific measurements) one ought to converge to a very small solution set with irreducible error – the inherent epistemic uncertainty of the world. As more models are trained on internet-scale data, the set of solutions that satisfies all data constraints must become relatively small.

In addition to data-scaling, many modern representation learning objectives $\mathcal{L}(f,x)$ directly optimize for multi-task solving. Contrastive learning finds a distance structure over data samples that optimizes many classification tasks (Arora et al., 2019b; Wang & Isola, 2020; Tian et al., 2020b). Masked Autoencoders (He et al., 2021) optimize randomly sampled reconstruction tasks. In fact, autoregressive language modeling can also be seen as optimizing a diverse set of tasks (Radford et al., 2019). Such multi-task objectives may be more effective than single-task ones (*e.g.*, ImageNet classification) due to the fact that they impose more task constraints on the representation, leading to a smaller and higher-quality solution space (Chen et al., 2020; He et al., 2020; Radford et al., 2017; 2019).

3.2. Convergence via Model Capacity

Suppose there is a globally optimal representation for standard learning objectives. Then, under sufficient data, *scaling* a model (*i.e.*, using larger function classes $\mathcal{F}$), as well as improved optimization, should be more effective at finding better approximations to this optimum, as illustrated in Figure 5. With the same training objective, larger models, even of different architectures, will thus tend to converge toward this optimum. When different training objectives share similar minimizers, larger models are better at finding these minimizers, and will train to similar solutions over the training tasks. We summarize this hypothesis as follows: